

Zain Khan

software / mechatronics engineer



[zainmk](#)



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[zainkhan.vercel](#)

work experience

September 2024 - Present

Calgary, AB



GCT Maintenance Solutions | Software Engineer

- Lead developer for a SaaS ERP platform built for the mining industry. Full stack development from the ReactJS frontend; to the RESTful NodeJS backend API server.
- Knex & SQL for schema design and database management, creating quick and efficient queries/mutations. Migration/Seed files for testing and database recovery plans.
- Establishing Azure Cloud infrastructure resources to facilitate application. Adhering to RBAC for authentication/authorization between services, users and developers. Terraform (IaC) to automate creating/managing infrastructure across multiple environments. Github Actions workflows to setup automating CI/CD flows across multiple environments.
- Frequent 'discovery' and PoC based projects to explore sustaining application development and iterating upon new emerging tools.
- Defining software design coding principles & practices. Leading a team of developers (interns) to effectively distribute, assign and complete tasks as per Agile principles.
- Developer, as contract, for AgnicoEagle's JDEO ETL rollout project; assisting with transitioning data from SAP ERP systems to JDE, for several Ontario-based mines.
- Planning, writing, documenting ETL flows for various resources with Pyspark (Databricks), SQL and authenticated Azure resources. Primarily responsible for transformation & filtering of data to meet business and schema requirements/specifications as well as loading data via orchestration APIs as defined per documentation.
- Auditing data points during transformation via assigned 'keys' to confirm ETL flows and documentation to explain ETL processes for final approval and handover to client.

August 2021 - June 2023

Toronto, ON



Heliolytics | Full Stack Developer - Applied ML

- Led development of a React-based internal analysis tool replacing legacy tooling for thermal image inspection — applying computer vision and ML algorithms to detect and classify deformities across solar panel arrays.
- Built and maintained full-stack infrastructure for the platform: GraphQL database design, Python-based ML/automation scripts, and internal API endpoints, with PyTest coverage to validate detection accuracy and reliability.
- Contributed to a SaaS satellite imaging product delivering automated visual assessments for solar array site monitoring and asset management.
- Defined cross-functional workflows between product, design, and QA to standardize how inspection data and findings were communicated across teams.
- Documented system architecture and technical decisions, translating computer vision and infrastructure design into accessible terms for non-technical stakeholders.



contact

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education

Western University | 2016 - 2021

- Mechatronics Systems Engineering (BESc.)
- Computer Science (BSc.)

</> technical stack

- **FRONTEND** | ReactJS, TypeScript, HTML, CSS
- **BACKEND** | Python, Java, NodeJS
- **AI / ML** | TensorFlow, Keras, PyTorch, Numpy, Pandas, Pillow, Cursor IDE
- **EMBEDDED** | C, C++, MATLAB, Simulink
- **DATABASE** | SQL, MySQL, GraphQL, PostgreSQL, MongoDB, Knex, Pyspark
- **CLOUD** | Unix, Git, Kubernetes, Docker, Azure Cloud, Terraform, Github Actions, Databricks

certificates

MICROSOFT CERTIFIED: AZURE FUNDAMENTALS



February 20th, 2025

IBM AI ENGINEERING BY IBM ON COURSERA



August 8th, 2024

May 2020 – August 2020
Calgary, AB



University of Calgary | *Developer*

- Primary developer as part of a multidisciplinary team to assist in transitioning teaching resources online for the Chemical Engineering Department at the University of Calgary, during COVID-19.
- Utilized JavaScript, HTML and CSS to create interactive web pages consisting of educational content and calculators for students to use and a database to keep track of individual progress.

extracurricular & projects

Java

LocalNeighborhoodCNN

- A locally implemented convolutional neural network, developed to understand the CNN architecture and how such a network operates to classify images efficiently. Trained and tested on the MNIST database.
- Programmed in Java from the ground up, including implementations for matrix operations, backpropagation calculus as well as the convolutional, max pool, and fully connected neural network layers.

C++, MATLAB

MSE DESIGN PROJECT

- Mechatronics design project focused on creating a fully autonomous robot to perform a series of coordinated tasks incorporating aspects of electrical, mechanical and software design principles
- Utilized an onboard Arduino microcontroller programmed with C++. Troubleshooting the multidisciplinary aspects of the design, with the use of tools such as an oscilloscope, multimeter, and Matlab's Simulink to model transducers.

ReactJS, Python

thumbs

- A web application allowing users to track 'stories' in the form of movies and games, while allowing ratings on a binary system. Utilizing the collected data to further predict and suggest new content.
- Based in ReactJS with Material UI components and a supporting FastAPI – Python backend to serve endpoints for managing computationally heavy and secure processes such as authorization and database security.

Python, C++

WERC (Western Engineering Rocketry Club)

- Responsible for managing and implementing onboard control systems to collect telemetry information during the rocket's flight.
- Utilized a Raspberry Pi with C++ to collect data such as altitude and speed, while utilizing Python scripts to filter the data to a more usable form for analysis to improve on future flights

Python

PyCOVID

- A script written to automatically collect COVID and environmental data to perform various ML analysis and predict trends of cases as well as possible influencing factors.
- Written in Python utilizing Selenium to automate web scraping publicly accessible data and sklearn to perform machine learning data analysis on the data collected.